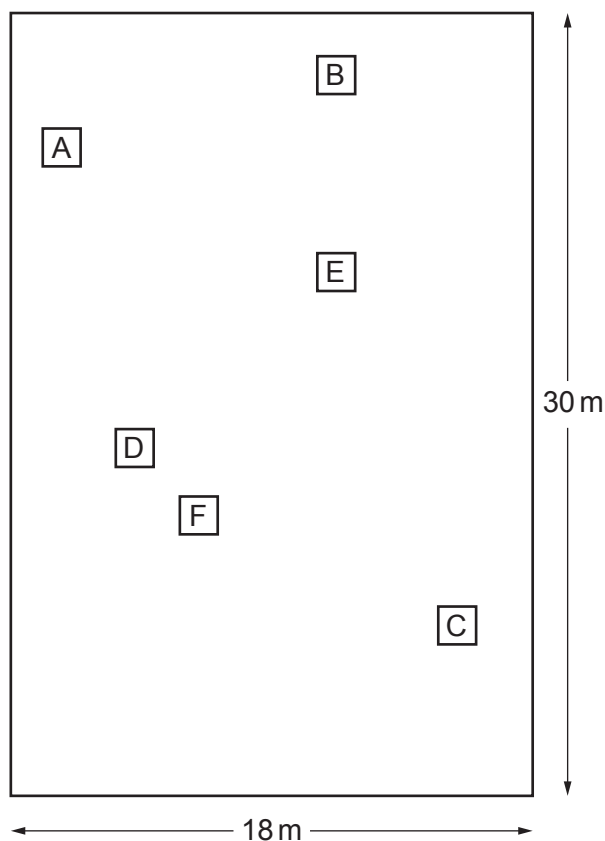


6. (a) Some students investigated the number of dandelion plants on a lawn. The diagram shows the lawn and the location of 6 quadrats (**A to F**) of area  $1 \text{ m}^2$  which the students had placed at random on the lawn.



The students counted the number of dandelions in each quadrat and recorded their results in the table below.

| Quadrat | Number of dandelions |
|---------|----------------------|
| A       | 6                    |
| B       | 3                    |
| C       | 2                    |
| D       | 7                    |
| E       | 4                    |
| F       | 5                    |



- (i) Use the information to estimate the total number of dandelions on the lawn. [3]

estimated number of dandelions = .....

- (ii) The actual number of dandelions on the lawn is 1950.  
Use the formula below to calculate the percentage error of your estimate above. [2]

$$\text{percentage error} = \frac{\text{estimated number of dandelions} - \text{actual number of dandelions}}{\text{actual number of dandelions}} \times 100$$

percentage error = .....

- (b) A species of dandelion has become resistant to one type of weedkiller. Explain, in terms of natural selection, how this has occurred. [4]

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